DERIVED LOWER SHRIEK COMMUTES WITH DIRECT SUMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Adjointness of j_! and restriction (Cohomology on Sites, Lemma sites-cohomology-lemma-adjoint-lower-shriek-restrict, canonical tag 08FJ). cited at proof lines 3779-3781. j_! for an open immersion is left adjoint to pullback and hence commutes with direct sums.
- Direct sums and bounded below pushforward (Etale Cohomology, Lemma etale-cohomology-lemma-direct-sum-bounded-below-pushforward, canonical tag 0GIW). cited at proof lines 3782-3785. Handles the proper case for objects bounded below.
- Direct sums for pushforward with torsion coefficients (Etale Cohomology, Lemma etale-cohomology-lemma-proper-mod-n-direct-sums, canonical tag 0F0D). cited at proof lines 3785-3787. Handles the proper case when Lambda is a torsion ring.

Source reference. Original lemma 08FJ: sources/stacks-chow/tags.txt, source lines 10712–10712. Cited at proof lines 3779-3781: j_! for an open immersion is left adjoint to pullback and hence commutes with direct sums.

Source reference. Original lemma 0GIW: sources/stacks-chow/tags.txt, source lines 20631–20631. Cited at proof lines 3782-3785: handles the proper case for objects bounded below.

Source reference. Original lemma 0F0D: sources/stacks-chow/tags.txt, source lines 18758–18758. Cited at proof lines 3785-3787: handles the proper case when Lambda is a torsion ring.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. Let Λ be a ring.*

(1) *Let $K_i \in D^+_{tors}(X_{étale}, \Lambda)$, $i \in I$ be a family of objects. Assume given $a \in \mathbf{Z}$ such that $H^n(K_i) = 0$ for $n < a$ and $i \in I$. Then $Rf_!(\bigoplus_i K_i) = \bigoplus_i Rf_!K_i$.*
(2) *If Λ is torsion, then the functor $Rf_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ commutes with direct sums.*

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Proof. By construction it suffices to prove this when f is an open immersion and when f is a proper morphism. For any open immersion $j : U \to X$ of schemes, the functor $j_! : D(U_{étale}) \to D(X_{étale})$ is a left adjoint to pullback $j^{-1} : D(X_{étale}) \to D(U_{étale})$ and hence commutes with direct sums, see Cohomology on Sites, Lemma 08FJ. In the proper case we have $Rf_! = Rf_*$ and we get the result from Étale Cohomology, Lemma 0GIW in the bounded belo case and from Étale Cohomology, Lemma 0F0D in the case that our coefficient ring Λ is a torsion ring. □